CS23334 Fundamentals of Data Science Lab

Experiment 5

Data Collection and Initial Exploration

Objective: To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

- Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)
- Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

1. Data Collection:

```
import pandas as pd

diabetes_df = pd.read_csv("diabetes.csv")

print(diabetes_df.head())
```

OUTPUT:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

2. Initial Exploration:

```
print(db.info())
print(db.describe())

import matplotlib.pyplot as plt
import seaborn as sns

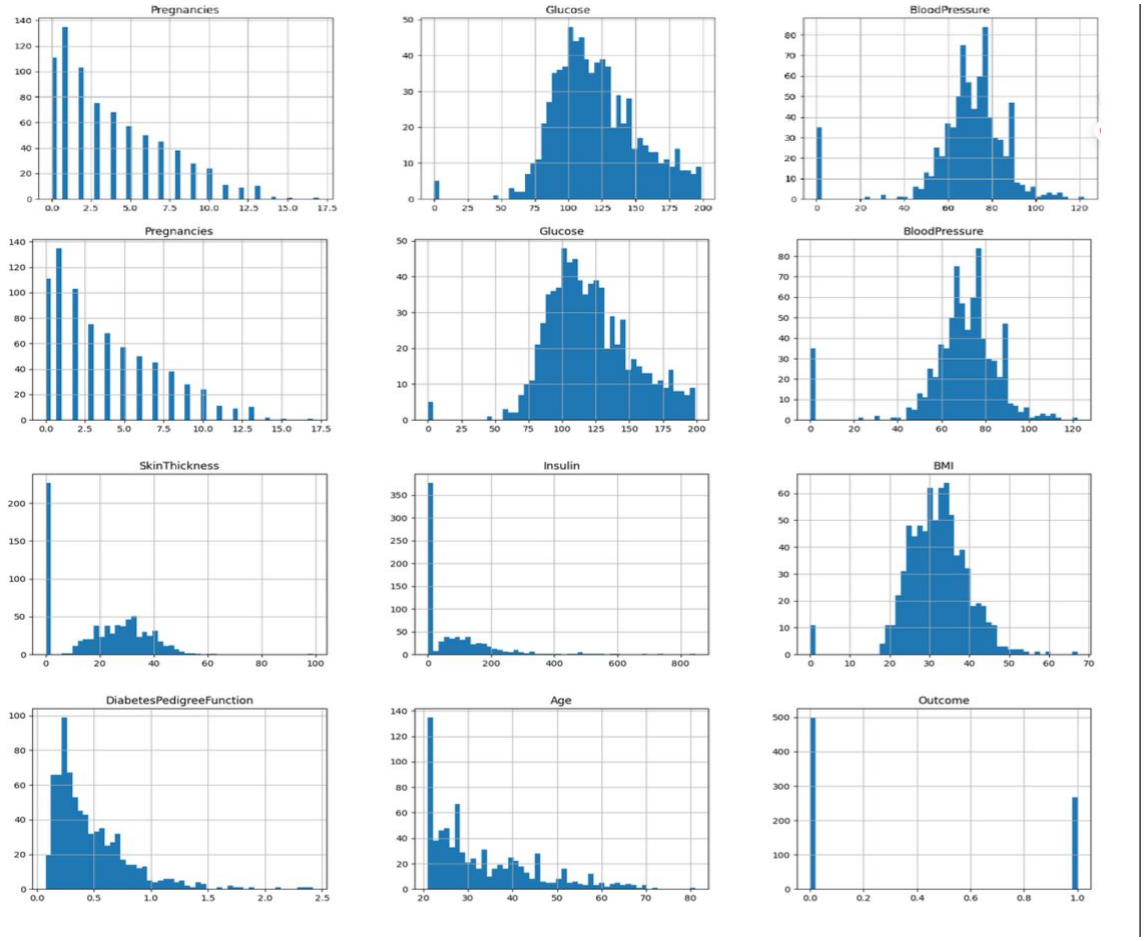
db.hist(bins=50, figsize=(20, 15))
plt.show()

sns.pairplot(db)
plt.show()
```

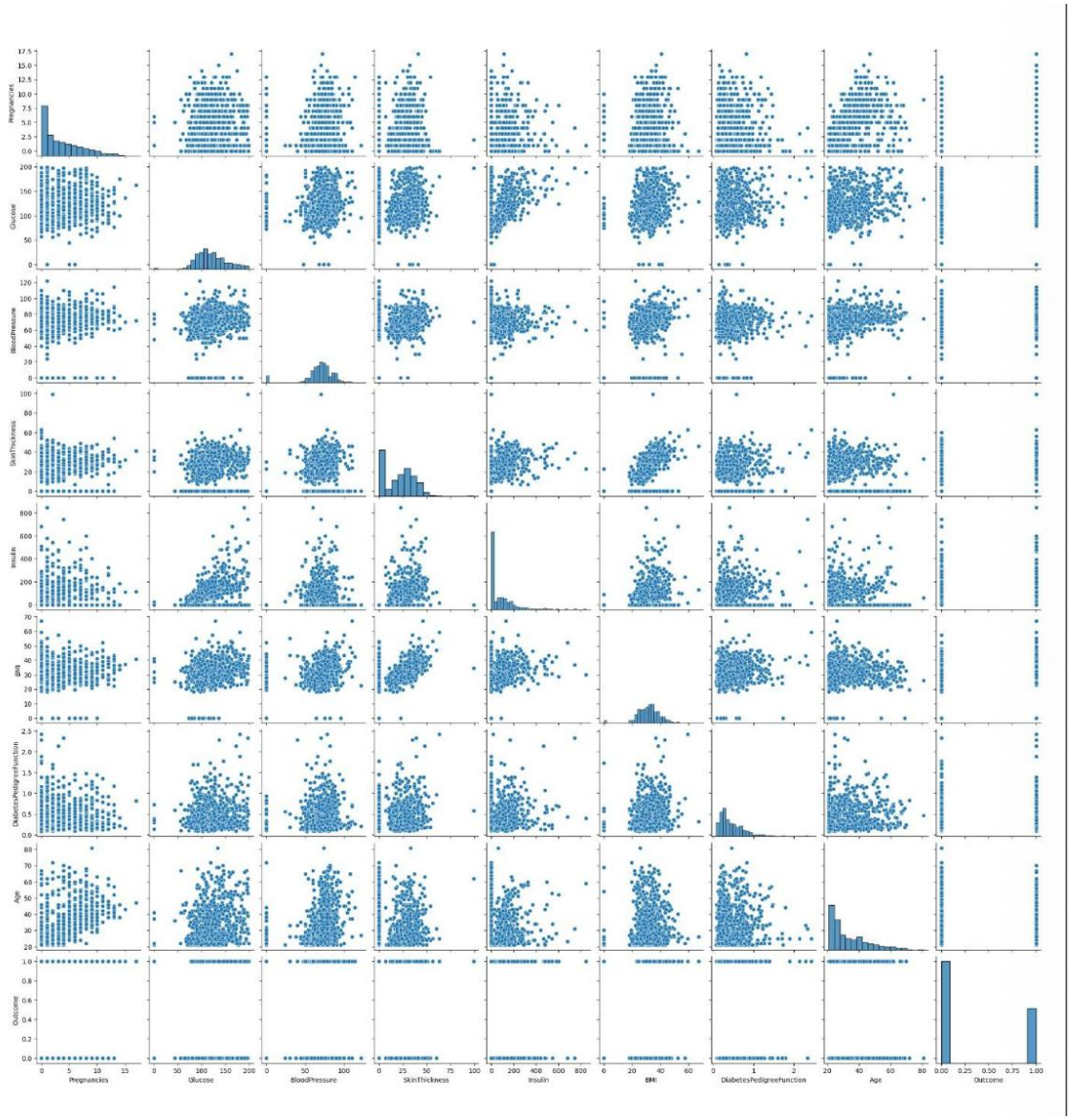
OUTPUT:

```
[3]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.845052  120.894531      69.105469      20.536458   79.799479
std       3.369578   31.972618      19.355807      15.952218  115.244002
min       0.000000    0.000000       0.000000       0.000000    0.000000
25%       1.000000   99.000000      62.000000       0.000000    0.000000
50%       3.000000  117.000000      72.000000      23.000000   30.500000
75%       6.000000  140.250000      80.000000      32.000000  127.250000
max      17.000000  199.000000     122.000000      99.000000  846.000000
```

OUTPUT:


Pregnancies
Glucose
BloodPressure
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome

OUTPUT:



Result:

The diabetes dataset was successfully loaded and explored. Summary statistics revealed key patterns such as higher glucose and BMI levels in diabetic patients. Distribution plots confirmed class imbalance with approximately 35% positive cases.